Homework 1 - Multivariate Analysis

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1. Exploratory Data Analysis

1a. Data Import

```
# Import dataset
data <- read.csv2("euroleague_25_26.csv")
# Remove row number
data$No <- NULL
# Inspect data
head(data)
```

```
##         TEAM                 PLAYER POSITION GP GS      Min PTS  X2P. X3P.  FT.
## 1 OLYMPIAKOS          THOMAS WALKUP    Guard 37 37 24:01:00 5.3  47.2 36.7 76.9
## 2 OLYMPIAKOS         FRANK NTILIKNA    Guard 24  6 16:24:00 4.5  40.0 36.5 84.2
## 3 OLYMPIAKOS           KEENAN EVANS    Guard  1  0  2:01:00 0.0   0.0  0.0  0.0
## 4 OLYMPIAKOS             TYSON WARD  Forward 35 14 19:05:00 7.1  56.7 26.4 68.4
## 5 OLYMPIAKOS GIANNOULIS LARENTZAKIS    Guard  7  0  2:59:00 0.3 100.0  0.0  0.0
## 6 OLYMPIAKOS             SABEN LEE    Guard 11  0 11:24:00 3.4  44.4 36.4 81.8
##    OR  DR  TR AST STL  TO BLK BLKA  FC  FD PIR
## 1 0.6 2.0 2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5
## 2 0.5 0.6 1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6
## 3 0.0 0.0 0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0
## 4 0.7 2.2 2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8
## 5 0.0 0.1 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0
## 6 0.0 0.4 0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8
```

```
str(data)
```

```
## 'data.frame':    71 obs. of  21 variables:
##  $ TEAM    : chr  "OLYMPIAKOS" "OLYMPIAKOS" "OLYMPIAKOS" "OLYMPIAKOS" ...
##  $ PLAYER  : chr  "THOMAS WALKUP" "FRANK NTILIKNA" "KEENAN EVANS" "TYSON WARD" ...
##  $ POSITION: chr  "Guard" "Guard" "Guard" "Forward" ...
##  $ GP      : int  37 24 1 35 7 11 2 8 39 36 ...
##  $ GS      : int  37 6 0 14 0 0 0 0 39 21 ...
##  $ Min     : chr  "24:01:00" "16:24:00" "2:01:00" "19:05:00" ...
##  $ PTS     : num  5.3 4.5 0 7.1 0.3 3.4 0 1.6 19 2.9 ...
##  $ X2P.    : num  47.2 40 0 56.7 100 44.4 0 30 61.8 64.3 ...
##  $ X3P.    : num  36.7 36.5 0 26.4 0 36.4 0 0 40.5 31.1 ...
##  $ FT.     : num  76.9 84.2 0 68.4 0 81.8 0 58.3 89 28.6 ...
##  $ OR      : num  0.6 0.5 0 0.7 0 0 0.5 0.2 1.4 0.3 ...
##  $ DR      : num  2 0.6 0 2.2 0.1 0.4 1 1 5 1.4 ...
##  $ TR      : num  2.6 1.1 0 2.9 0.1 0.4 1.5 1.2 6.4 1.8 ...
##  $ AST     : num  5.7 1.6 0 2.3 0.9 1.4 2.5 1.4 1.3 1.6 ...
##  $ STL     : num  0.9 0.5 0 0.5 0 0.4 0 0.1 0.6 0.4 ...
##  $ TO      : num  1.5 1.2 0 1.1 0.3 1.3 0 0.5 1.1 1 ...
```

```
##  $ BLK     : num  0 0.4 0 0.1 0 0.1 1 0.1 0.1 0.1 ...
##  $ BLKA    : num  0.1 0.2 0 0.3 0 0.2 0 0.2 0.2 0.1 ...
##  $ FC      : num  2.4 2.9 0 2.4 0.6 1.2 0 1 0.9 1.7 ...
##  $ FD      : num  1.8 1.4 0 2.3 0.3 0.6 0.5 0.9 3.1 0.3 ...
##  $ PIR     : num  9.5 2.6 0 7.8 0 1.8 5.5 1.2 22.1 2.3 ...
```

```
head(data)
```

```
##         TEAM                 PLAYER POSITION GP GS      Min PTS  X2P. X3P.  FT.
## 1 OLYMPIAKOS          THOMAS WALKUP    Guard 37 37 24:01:00 5.3  47.2 36.7 76.9
## 2 OLYMPIAKOS         FRANK NTILIKNA    Guard 24  6 16:24:00 4.5  40.0 36.5 84.2
## 3 OLYMPIAKOS           KEENAN EVANS    Guard  1  0  2:01:00 0.0   0.0  0.0  0.0
## 4 OLYMPIAKOS             TYSON WARD  Forward 35 14 19:05:00 7.1  56.7 26.4 68.4
## 5 OLYMPIAKOS GIANNOULIS LARENTZAKIS    Guard  7  0  2:59:00 0.3 100.0  0.0  0.0
## 6 OLYMPIAKOS             SABEN LEE    Guard 11  0 11:24:00 3.4  44.4 36.4 81.8
##    OR  DR  TR AST STL  TO BLK BLKA  FC  FD PIR
## 1 0.6 2.0 2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5
## 2 0.5 0.6 1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6
## 3 0.0 0.0 0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0
## 4 0.7 2.2 2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8
## 5 0.0 0.1 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0
## 6 0.0 0.4 0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8
```

1b. Convert Minutes

```
# Split minutes
aux <- strsplit(data$Min, ":")

# Extract minutes and convert to numeric
data$`Min 2` <- as.numeric(sapply(aux, `[`, 1))

# Inspect result
head(data[, c("Min", "Min 2")])
```

```
##        Min Min 2
## 1 24:01:00    24
## 2 16:24:00    16
## 3  2:01:00     2
## 4 19:05:00    19
## 5  2:59:00     2
## 6 11:24:00    11
```

1c. Set Player Names as Row Labels

```
# Assign player names as row labels
rownames(data) <- make.unique(as.character(data$PLAYER))

# Remove PLAYER and Min columns
data$PLAYER <- NULL
data$Min <- NULL

# Inspect result
head(data)
```

```
##                          TEAM POSITION GP GS PTS  X2P. X3P.  FT.  OR  DR
```

```
## THOMAS WALKUP          OLYMPIAKOS    Guard 37 37 5.3  47.2 36.7 76.9 0.6 2.0
## FRANK NTILIKNA         OLYMPIAKOS    Guard 24  6 4.5  40.0 36.5 84.2 0.5 0.6
## KEENAN EVANS           OLYMPIAKOS    Guard  1  0 0.0   0.0  0.0  0.0 0.0 0.0
## TYSON WARD             OLYMPIAKOS  Forward 35 14 7.1  56.7 26.4 68.4 0.7 2.2
## GIANNOULIS LARENTZAKIS OLYMPIAKOS    Guard  7  0 0.3 100.0  0.0  0.0 0.0 0.1
## SABEN LEE              OLYMPIAKOS    Guard 11  0 3.4  44.4 36.4 81.8 0.0 0.4
##                         TR AST STL  TO BLK BLKA  FC  FD PIR Min 2
## THOMAS WALKUP          2.6 5.7 0.9 1.5 0.0  0.1 2.4 1.8 9.5    24
## FRANK NTILIKNA         1.1 1.6 0.5 1.2 0.4  0.2 2.9 1.4 2.6    16
## KEENAN EVANS           0.0 0.0 0.0 0.0 0.0  0.0 0.0 0.0 0.0     2
## TYSON WARD             2.9 2.3 0.5 1.1 0.1  0.3 2.4 2.3 7.8    19
## GIANNOULIS LARENTZAKIS 0.1 0.9 0.0 0.3 0.0  0.0 0.6 0.3 0.0     2
## SABEN LEE              0.4 1.4 0.4 1.3 0.1  0.2 1.2 0.6 1.8    11
```

1d. Filter Players

```
# Number of players before filtering
nrow(data)
```

```
## [1] 71
```

```
# Keep players with at least 5 minutes and 5 games
data <- data[data$`Min 2` >= 5 & data$GP >= 5, ]

# Number of players after filtering
nrow(data)
```

```
## [1] 60
```

```
# Inspect result
head(data)
```

```
##                      TEAM POSITION GP GS  PTS X2P. X3P.  FT.  OR  DR  TR AST
## THOMAS WALKUP  OLYMPIAKOS    Guard 37 37  5.3 47.2 36.7 76.9 0.6 2.0 2.6 5.7
## FRANK NTILIKNA OLYMPIAKOS    Guard 24  6  4.5 40.0 36.5 84.2 0.5 0.6 1.1 1.6
## TYSON WARD     OLYMPIAKOS  Forward 35 14  7.1 56.7 26.4 68.4 0.7 2.2 2.9 2.3
## SABEN LEE      OLYMPIAKOS    Guard 11  0  3.4 44.4 36.4 81.8 0.0 0.4 0.4 1.4
## MONTE MORRIS   OLYMPIAKOS    Guard  8  0  1.6 30.0  0.0 58.3 0.2 1.0 1.2 1.4
## SASHA VEZENKOV OLYMPIAKOS  Forward 39 39 19.0 61.8 40.5 89.0 1.4 5.0 6.4 1.3
##                STL  TO BLK BLKA  FC  FD  PIR Min 2
## THOMAS WALKUP  0.9 1.5 0.0  0.1 2.4 1.8  9.5    24
## FRANK NTILIKNA 0.5 1.2 0.4  0.2 2.9 1.4  2.6    16
## TYSON WARD     0.5 1.1 0.1  0.3 2.4 2.3  7.8    19
## SABEN LEE      0.4 1.3 0.1  0.2 1.2 0.6  1.8    11
## MONTE MORRIS   0.1 0.5 0.1  0.2 1.0 0.9  1.2    11
## SASHA VEZENKOV 0.6 1.1 0.1  0.2 0.9 3.1 22.1    27
```

2. Principal Component Analysis (PCA)

2a

```
#install.packages("FactoMineR")

library(FactoMineR)
```

```
## Warning: Paket 'FactoMineR' wurde unter R Version 4.4.3 erstellt
```

```r
# Convert categorical variables to factors
data$TEAM <- as.factor(data$TEAM)
data$POSITION <- as.factor(data$POSITION)

# Identify supplementary variables
quali_vars <- which(names(data) %in% c("TEAM", "POSITION"))
quanti_vars <- which(names(data) == "PIR")

# Apply PCA
pca <- PCA(
  data,
  scale.unit = TRUE,
  quali.sup = quali_vars,
  quanti.sup = quanti_vars,
  graph = FALSE
)

# Inspect PCA results
summary(pca)
```

```
##
## Call:
## PCA(X = data, scale.unit = TRUE, quanti.sup = quanti_vars, quali.sup = quali_vars,
##      graph = FALSE)
##
##
## Eigenvalues
##                        Dim.1   Dim.2   Dim.3   Dim.4   Dim.5   Dim.6   Dim.7
## Variance               6.681   3.278   1.319   1.222   0.837   0.761   0.746
## % of var.             39.302  19.285   7.759   7.188   4.924   4.477   4.390
## Cumulative % of var.  39.302  58.587  66.346  73.534  78.458  82.935  87.325
##                        Dim.8   Dim.9  Dim.10  Dim.11  Dim.12  Dim.13  Dim.14
## Variance               0.540   0.428   0.306   0.294   0.211   0.154   0.096
## % of var.              3.176   2.518   1.798   1.732   1.244   0.903   0.567
## Cumulative % of var.  90.501  93.019  94.816  96.548  97.792  98.695  99.262
##                       Dim.15  Dim.16  Dim.17
## Variance               0.078   0.047   0.000
## % of var.              0.461   0.275   0.002
## Cumulative % of var.  99.723  99.998 100.000
##
## Individuals (the 10 first)
##                        Dist    Dim.1    ctr   cos2    Dim.2    ctr   cos2
## THOMAS WALKUP      |  4.651 |  1.845  0.849  0.157 |  2.568  3.352  0.305 |
## FRANK NTILIKNA     |  3.202 | -1.345  0.451  0.176 |  1.163  0.687  0.132 |
## TYSON WARD         |  1.832 |  0.807  0.163  0.194 |  0.342  0.059  0.035 |
## SABEN LEE          |  4.053 | -3.282  2.686  0.656 |  1.753  1.562  0.187 |
## MONTE MORRIS       |  5.268 | -4.171  4.339  0.627 | -0.092  0.004  0.000 |
## SASHA VEZENKOV     |  6.218 |  4.716  5.548  0.575 | -0.355  0.064  0.003 |
## KOSTAS PAPANIKOLAU |  4.201 | -1.676  0.701  0.159 | -0.759  0.293  0.033 |
## TYLER DORSEY       |  5.155 |  3.567  3.173  0.479 |  2.836  4.088  0.303 |
## ALEC PETERS        |  2.435 | -0.118  0.003  0.002 | -0.075  0.003  0.001 |
## NIKOLA MILUTINOV   |  7.115 |  4.382  4.791  0.379 | -3.960  7.974  0.310 |
##                     Dim.3    ctr   cos2
```

```
## THOMAS WALKUP       0.739  0.691  0.025 |
## FRANK NTILIKNA     -0.968  1.185  0.091 |
## TYSON WARD         -0.491  0.305  0.072 |
## SABEN LEE          -0.720  0.655  0.032 |
## MONTE MORRIS       -1.859  4.369  0.125 |
## SASHA VEZENKOV      1.489  2.802  0.057 |
## KOSTAS PAPANIKOLAU  1.403  2.489  0.112 |
## TYLER DORSEY       -0.782  0.773  0.023 |
## ALEC PETERS         1.436  2.607  0.348 |
## NIKOLA MILUTINOV   -0.358  0.162  0.003 |
##
## Variables (the 10 first)
##                       Dim.1    ctr   cos2    Dim.2    ctr   cos2    Dim.3
## GP                  |  0.703  7.406  0.495 | -0.022  0.015  0.000 |  0.467
## GS                  |  0.732  8.030  0.537 | -0.001  0.000  0.000 |  0.231
## PTS                 |  0.857 11.002  0.735 |  0.296  2.674  0.088 | -0.022
## X2P.                |  0.481  3.459  0.231 | -0.538  8.844  0.290 |  0.187
## X3P.                |  0.050  0.038  0.003 |  0.561  9.592  0.314 |  0.630
## FT.                 |  0.191  0.544  0.036 |  0.418  5.326  0.175 | -0.083
## OR                  |  0.577  4.990  0.333 | -0.732 16.352  0.536 | -0.062
## DR                  |  0.817  9.989  0.667 | -0.380  4.402  0.144 |  0.122
## TR                  |  0.788  9.300  0.621 | -0.542  8.975  0.294 |  0.063
## AST                 |  0.475  3.383  0.226 |  0.702 15.021  0.492 | -0.091
##                       ctr   cos2
## GP                 16.508  0.218 |
## GS                  4.033  0.053 |
## PTS                 0.037  0.000 |
## X2P.                2.661  0.035 |
## X3P.               30.128  0.397 |
## FT.                 0.527  0.007 |
## OR                  0.290  0.004 |
## DR                  1.133  0.015 |
## TR                  0.305  0.004 |
## AST                 0.631  0.008 |
##
## Supplementary continuous variable
##                       Dim.1   cos2    Dim.2   cos2    Dim.3   cos2
## PIR                 |  0.921  0.849 |  0.054  0.003 | -0.022  0.000 |
##
## Supplementary categories
##                        Dist    Dim.1   cos2 v.test    Dim.2   cos2 v.test
## FENERBAHCE          |  0.844 | -0.573  0.460 -1.070 |  0.222  0.069  0.592 |
## OLYMPIAKOS          |  0.620 |  0.242  0.152  0.414 |  0.054  0.008  0.132 |
## REAL MADRID         |  0.582 |  0.245  0.177  0.402 | -0.239  0.168 -0.559 |
## VALENCIA BASKET     |  0.537 |  0.192  0.128  0.314 | -0.088  0.027 -0.207 |
## Center              |  2.163 | -0.055  0.001 -0.099 | -2.110  0.951 -5.398 |
## Forward             |  1.472 |  1.149  0.609  2.236 |  0.092  0.004  0.254 |
## Guard               |  1.509 | -0.762  0.255 -1.980 |  1.235  0.670  4.582 |
##                       Dim.3   cos2 v.test
## FENERBAHCE         -0.093  0.012 -0.390 |
## OLYMPIAKOS         -0.124  0.040 -0.481 |
## REAL MADRID         0.030  0.003  0.110 |
## VALENCIA BASKET     0.216  0.163  0.798 |
## Center             -0.348  0.026 -1.404 |
```

```
## Forward             0.661  0.202  2.895 |
## Guard              -0.244  0.026 -1.425 |
```

2b. Number of Principal Components

```r
# Display eigenvalues and explained variance
pca$eig
```

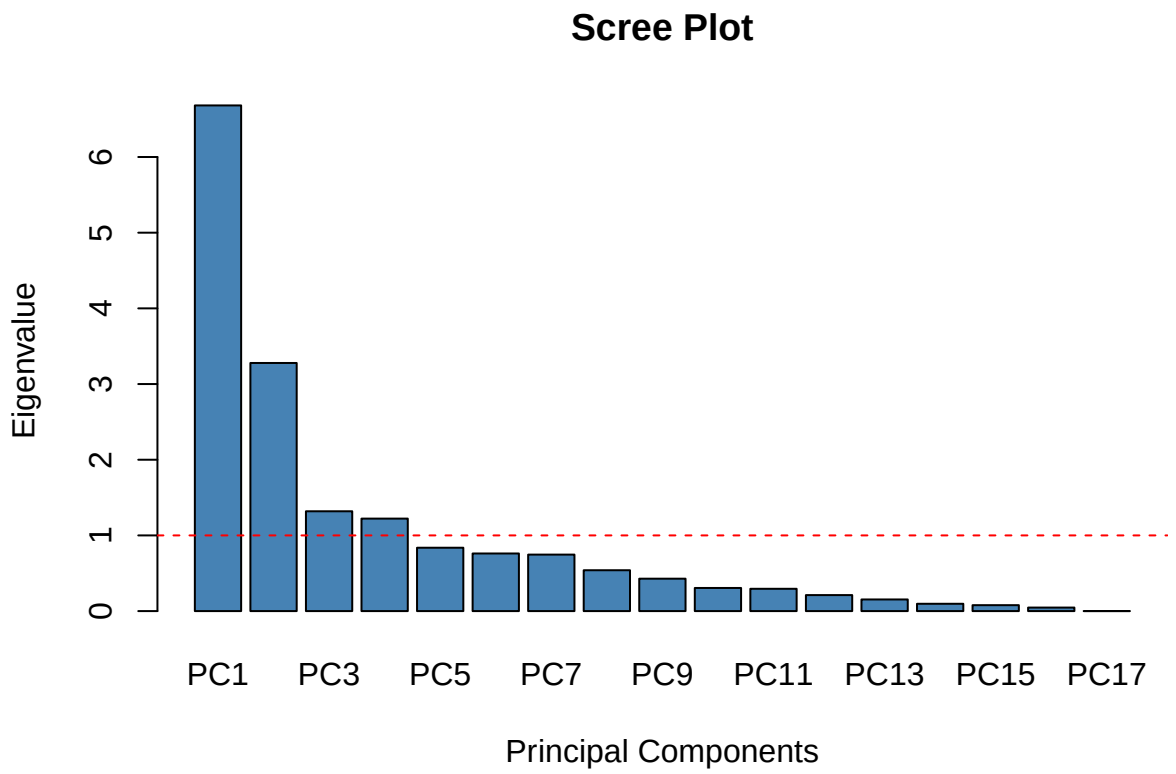
```
##           eigenvalue percentage of variance cumulative percentage of variance
## comp 1  6.6813634639           39.302138023                           39.30214
## comp 2  3.2784134746           19.284785145                           58.58692
## comp 3  1.3189846363            7.758733155                           66.34566
## comp 4  1.2219880847            7.188165204                           73.53382
## comp 5  0.8371559249            4.924446617                           78.45827
## comp 6  0.7610571829            4.476806958                           82.93508
## comp 7  0.7462893365            4.389937273                           87.32501
## comp 8  0.5398916603            3.175833296                           90.50085
## comp 9  0.4280176133            2.517750666                           93.01860
## comp 10 0.3056429104            1.797899473                           94.81650
## comp 11 0.2944111859            1.731830505                           96.54833
## comp 12 0.2114119624            1.243599779                           97.79193
## comp 13 0.1535485815            0.903226950                           98.69515
## comp 14 0.0964251760            0.567206918                           99.26236
## comp 15 0.0783187980            0.460698812                           99.72306
## comp 16 0.0466737778            0.274551634                           99.99761
## comp 17 0.0004062307            0.002389592                          100.00000
```

```r
# Number of components with eigenvalue > 1
sum(pca$eig[, 1] > 1)
```

```
## [1] 4
```

```r
# Scree plot
barplot(
  pca$eig[, 1],
  names.arg = paste0("PC", 1:nrow(pca$eig)),
  xlab = "Principal Components",
  ylab = "Eigenvalue",
  main = "Scree Plot",
  col = "steelblue"
)

abline(h = 1, col = "red", lty = 2)
```



Based on the Kaiser criterion, four principal components should be retained, as their eigenvalues exceed 1. The scree plot shows a particularly steep decline after PC2, indicating that the first two components explain most of the variance.

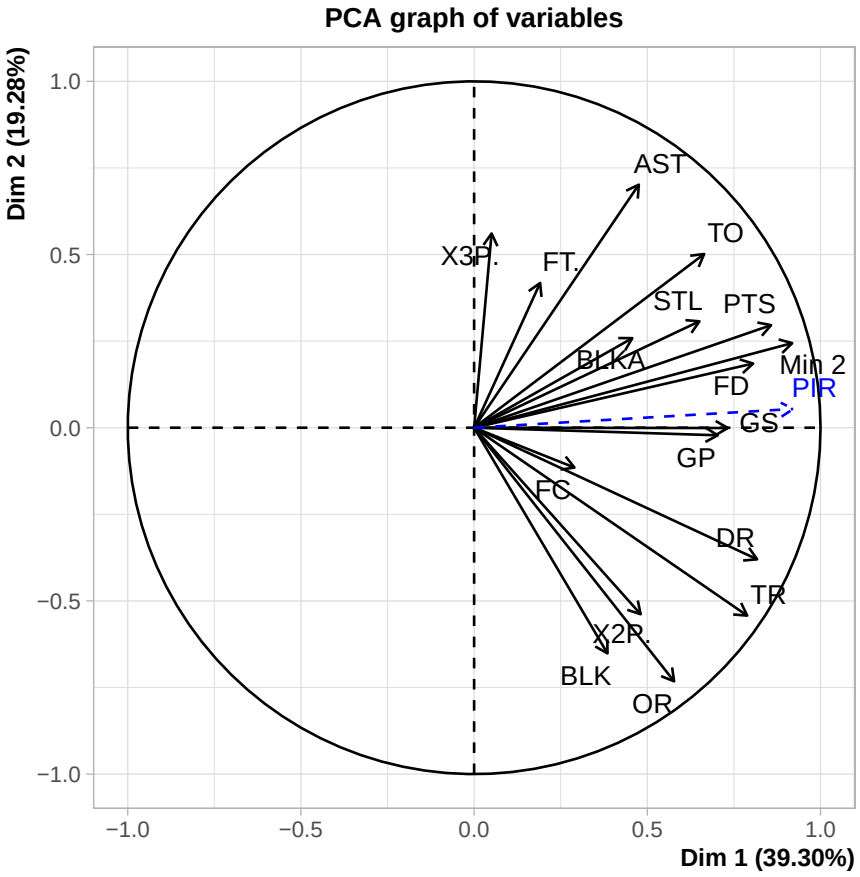
2c

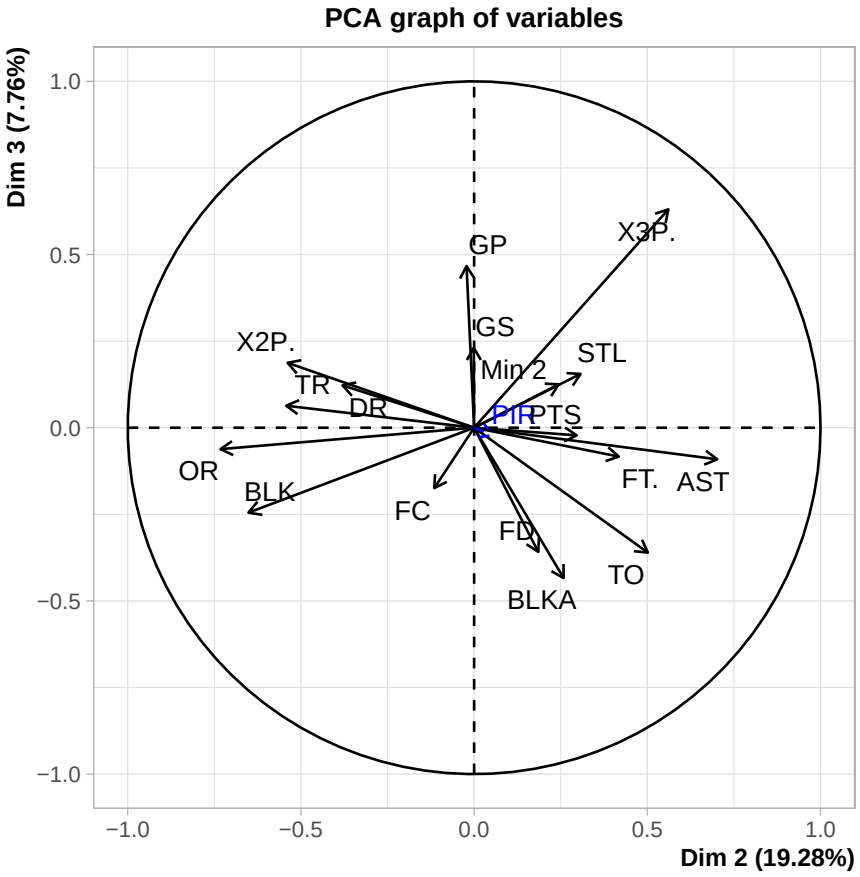
```
##            Dim.1         Dim.2       Dim.3        Dim.4        Dim.5
## GP     0.70345944 -0.0218546402  0.46662016 -0.056788043  0.219930804
## GS     0.73246819 -0.0009066644  0.23064869 -0.078071104  0.159535929
## PTS    0.85736119  0.2960999943 -0.02204878 -0.157100124 -0.018189268
## X2P.   0.48077204 -0.5384501095  0.18734335  0.245175958 -0.261173909
## X3P.   0.05039609  0.5607721596  0.63038821  0.070943172  0.250469683
## FT.    0.19057912  0.4178613912 -0.08333394 -0.578772875 -0.121321703
## OR     0.57741509 -0.7321762370 -0.06182125 -0.100781973 -0.037612249
## DR     0.81696083 -0.3798924520  0.12224829 -0.159523796  0.008739367
## TR     0.78827264 -0.5424339522  0.06342686 -0.145476847 -0.015638587
## AST    0.47544687  0.7017542494 -0.09120426  0.155127098 -0.256352578
## STL    0.65047380  0.3071519638  0.15536757  0.280322422 -0.327857982
## TO     0.66412520  0.5014585478 -0.36043627  0.205470148  0.029433507
## BLK    0.38495635 -0.6514179288 -0.24511167 -0.004224641  0.086962077
## BLKA   0.45663599  0.2585049001 -0.43397239 -0.211570921  0.533758547
## FC     0.28967268 -0.1150539616 -0.17437917  0.735470315  0.313240362
## FD     0.80535676  0.1852181747 -0.35843977 -0.005248291 -0.222665236
## Min 2  0.91830049  0.2442284079  0.12519550  0.022578337  0.036994593
```

```
##            Dim.1         Dim.2       Dim.3        Dim.4        Dim.5
## GP     0.70345944 -0.0218546402  0.46662016 -0.056788043  0.219930804
```

```
## GS    0.73246819 -0.0009066644  0.23064869 -0.078071104  0.159535929
## PTS   0.85736119  0.2960999943 -0.02204878 -0.157100124 -0.018189268
## X2P.  0.48077204 -0.5384501095  0.18734335  0.245175958 -0.261173909
## X3P.  0.05039609  0.5607721596  0.63038821  0.070943172  0.250469683
## FT.   0.19057912  0.4178613912 -0.08333394 -0.578772875 -0.121321703
## OR    0.57741509 -0.7321762370 -0.06182125 -0.100781973 -0.037612249
## DR    0.81696083 -0.3798924520  0.12224829 -0.159523796  0.008739367
## TR    0.78827264 -0.5424339522  0.06342686 -0.145476847 -0.015638587
## AST   0.47544687  0.7017542494 -0.09120426  0.155127098 -0.256352578
## STL   0.65047380  0.3071519638  0.15536757  0.280322422 -0.327857982
## TO    0.66412520  0.5014585478 -0.36043627  0.205470148  0.029433507
## BLK   0.38495635 -0.6514179288 -0.24511167 -0.004224641  0.086962077
## BLKA  0.45663599  0.2585049001 -0.43397239 -0.211570921  0.533758547
## FC    0.28967268 -0.1150539616 -0.17437917  0.735470315  0.313240362
## FD    0.80535676  0.1852181747 -0.35843977 -0.005248291 -0.222665236
## Min 2 0.91830049  0.2442284079  0.12519550  0.022578337  0.036994593
```

```
##             Dim.1        Dim.2       Dim.3        Dim.4        Dim.5
## GP     7.40649988 1.456879e-02 16.50772644  0.263904523  5.777843398
## GS     8.02994263 2.507433e-05  4.03331600  0.498785325  3.040259518
## PTS   11.00176959 2.674318e+00  0.03685778  2.019696372  0.039520649
## X2P.   3.45949980 8.843562e+00  2.66095055  4.919135548  8.148041349
## X3P.   0.03801269 9.592000e+00 30.12842490  0.411864385  7.493832395
## FT.    0.54360764 5.325995e+00  0.52650689 27.412545597  1.758209566
## OR     4.99012194 1.635187e+01  0.28975829  0.831186999  0.168986591
## DR     9.98935324 4.402077e+00  1.13304152  2.082495053  0.009123333
## TR     9.30010403 8.974908e+00  0.30500480  1.731891923  0.029213842
## AST    3.38328741 1.502126e+01  0.63065300  1.969284068  7.849988576
## STL    6.33278166 2.877682e+00  1.83012602  6.430558626 12.840004266
## TO     6.60138142 7.670194e+00  9.84956897  3.454860337  0.103485066
## BLK    2.21798123 1.294362e+01  4.55499857  0.001460537  0.903344602
## BLKA   3.12086634 2.038327e+00 14.27856190  3.663068026 34.031675339
## FC     1.25588532 4.037750e-01  2.30541695 44.265291234 11.720579315
## FD     9.70759217 1.046414e+00  9.74075572  0.002254077  5.922410119
## Min 2 12.62131301 1.819402e+00  1.18833170  0.041717370  0.163482076
```


PCA graph of variables
Dim 2 (19.28%)
Dim 1 (39.30%)
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
AST
TO
X3P.
FT.
STL
PTS
BLKA
Min 2
FD
PIR
GS
GP
FC
DR
TR
X2P.
BLK
OR

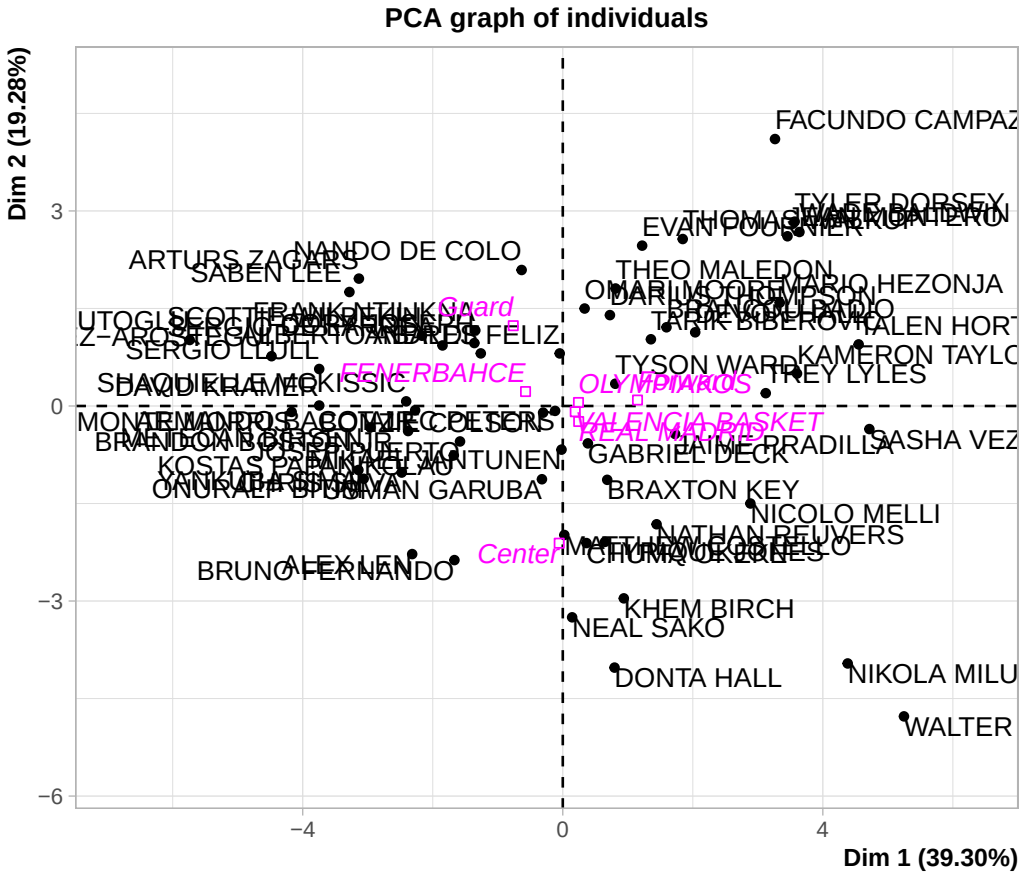


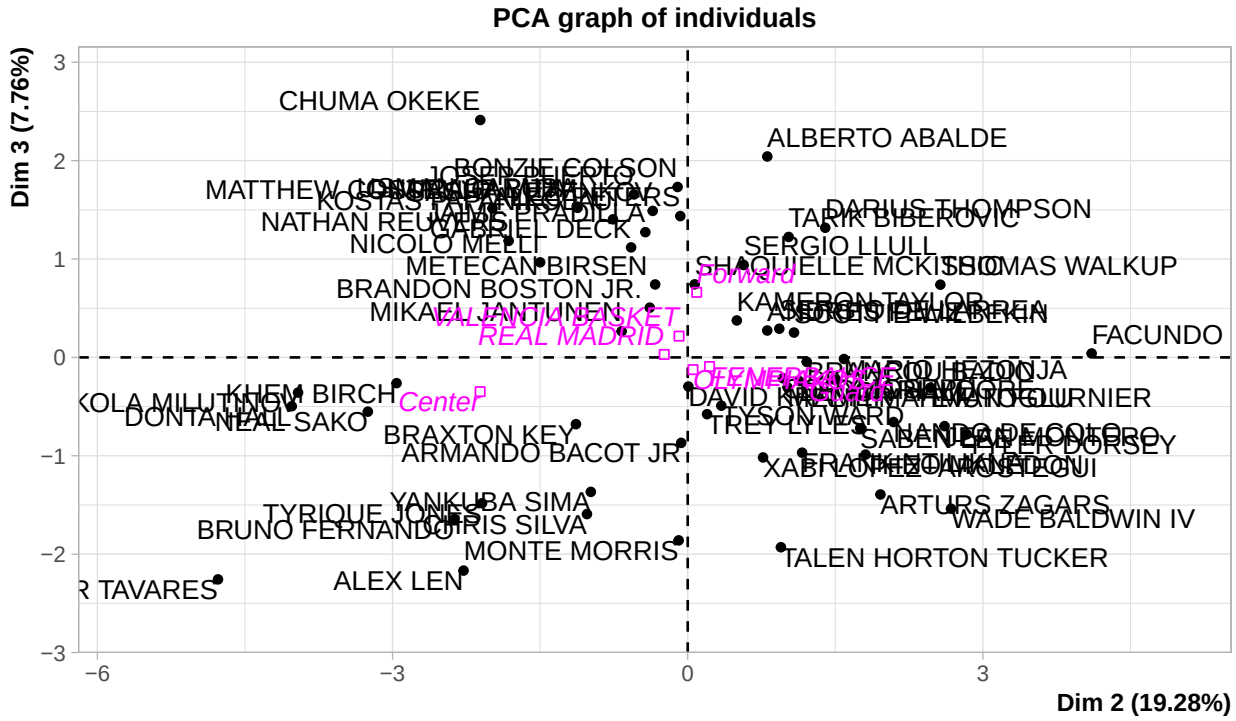
2d

PC1 (39.30%) – Overall Player Activity: Most statistical variables are positively correlated with PC1, particularly points, minutes played, and rebounds. This dimension represents overall player activity and statistical contribution.

PC2 (19.28%) – Playmaking vs. Rebounding: Assists, turnovers, and steals correlate positively with PC2, while rebounds and blocks correlate negatively. This dimension distinguishes playmaking-oriented players from inside players.

PC3 (7.76%) – Three-Point Shooting vs. Physical Play: Three-point shooting percentage and games played correlate positively with PC3, while blocks against and turnovers correlate negatively. This dimension may reflect differences in playing style, although its interpretation is less clear.


PCA graph of individuals
Dim 2 (19.28%)
Dim 1 (39.30%)
3
0
−3
−6
−4
0
4
FACUNDO CAMPAZ
TYLER DORSEY
EVAN FOURNIER
THEO MALEDON
NANDO DE COLO
ARTURS ZAGARS
SABEN LEE
OMARI MOORE
MARIO HEZONJA
Guard
TARIK BIBEROVIC
KALEN HORT
KAMERON TAYLO
TYSON WARD
TREY LYLES
FENERBAHCE
OLYMPIACOS
VALENCIA BASKET
REAL MADRID
GABRIEL DECK
SASHA VEZ
BRAXTON KEY
NICOLO MELLI
NATHAN REUVERS
Center
BRUNO FERNANDO
KHEM BIRCH
NEAL SAKO
DONTA HALL
NIKOLA MILU
WALTER

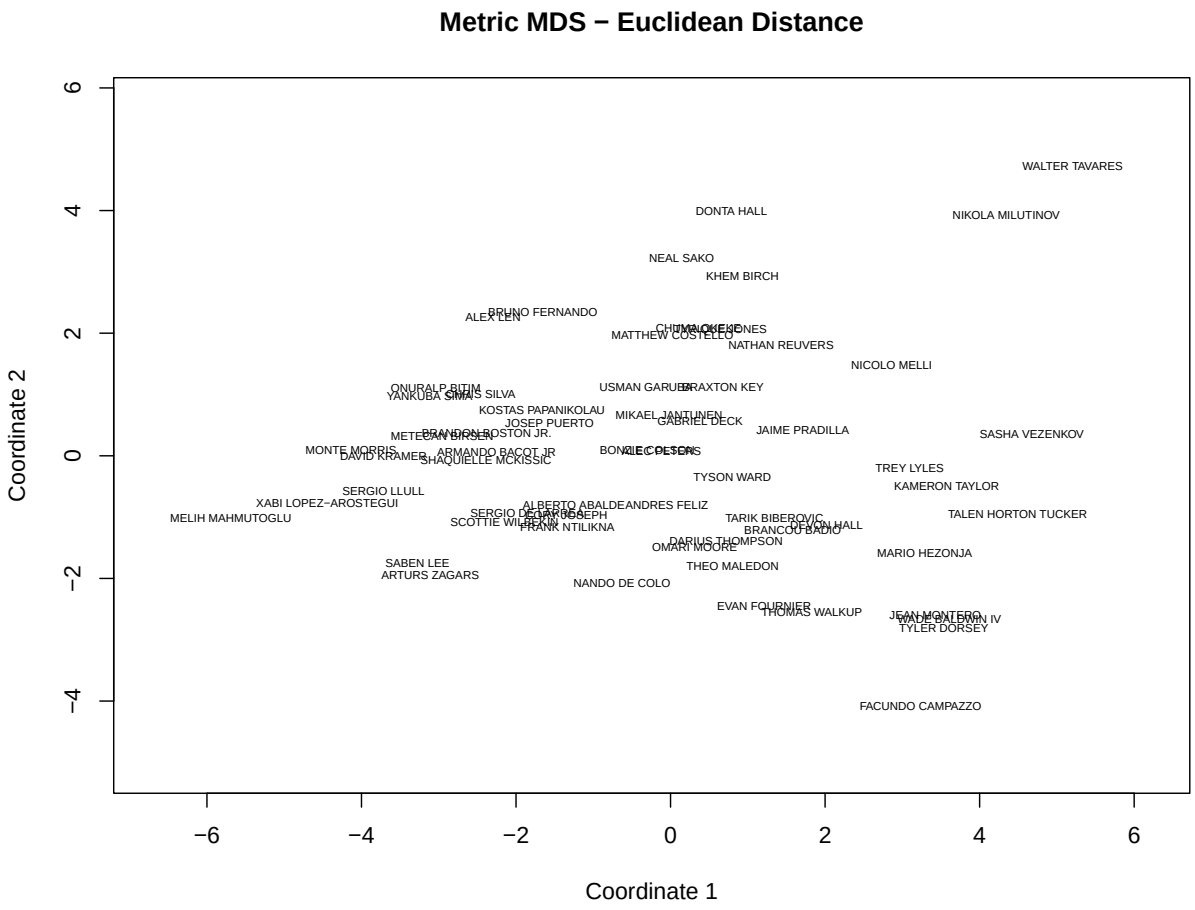


The PC1–PC2 plot (58.58% explained variance) separates players mainly by overall statistical activity (PC1) and playing style (PC2). Players 33 and 14 show high PC1 scores, while player 44 has a low score. PC2 distinguishes playmaking-oriented players, such as player 24, from more inside-oriented players, such as players 33 and 17.

The PC2–PC3 plot (27.04% explained variance) highlights additional differences in playing style, particularly three-point shooting. Players 25 and 23 have relatively high PC3 scores. Player positions partly reflect their roles, with guards tending towards higher PC2 values and centers towards lower values. However, the teams overlap considerably, indicating no clear separation by team.

3. Multidimensional Scaling (MDS)

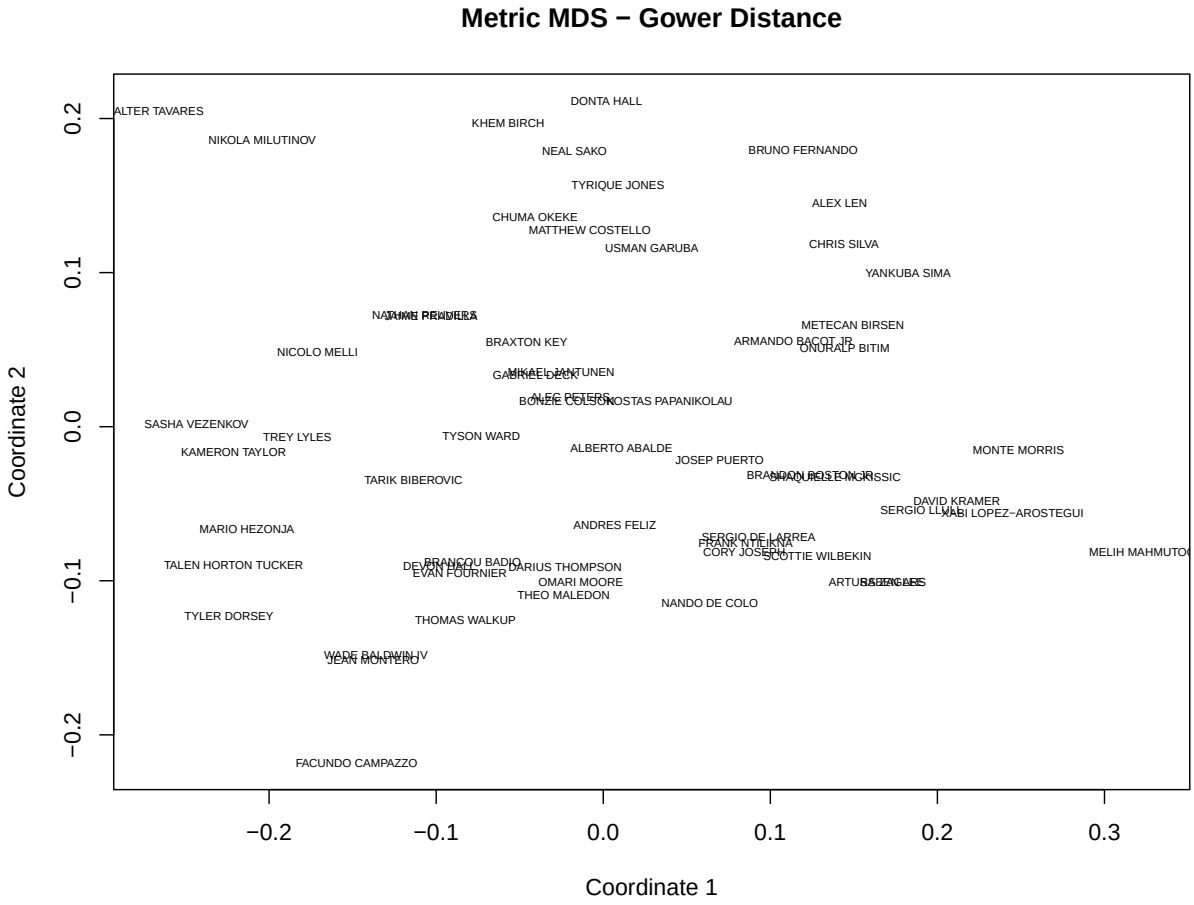
3a. Metric MDS - Euclidean Distance



3b

The Euclidean MDS plot shows a central concentration of players with relatively similar statistical profiles, without clearly separated clusters. Some players, such as Walter Tavares, Nikola Milutinov, and Facundo Campazzo, occupy more isolated positions, indicating distinct playing characteristics. Overall, the plot reveals both similarities among many players and substantial differences for certain individuals.

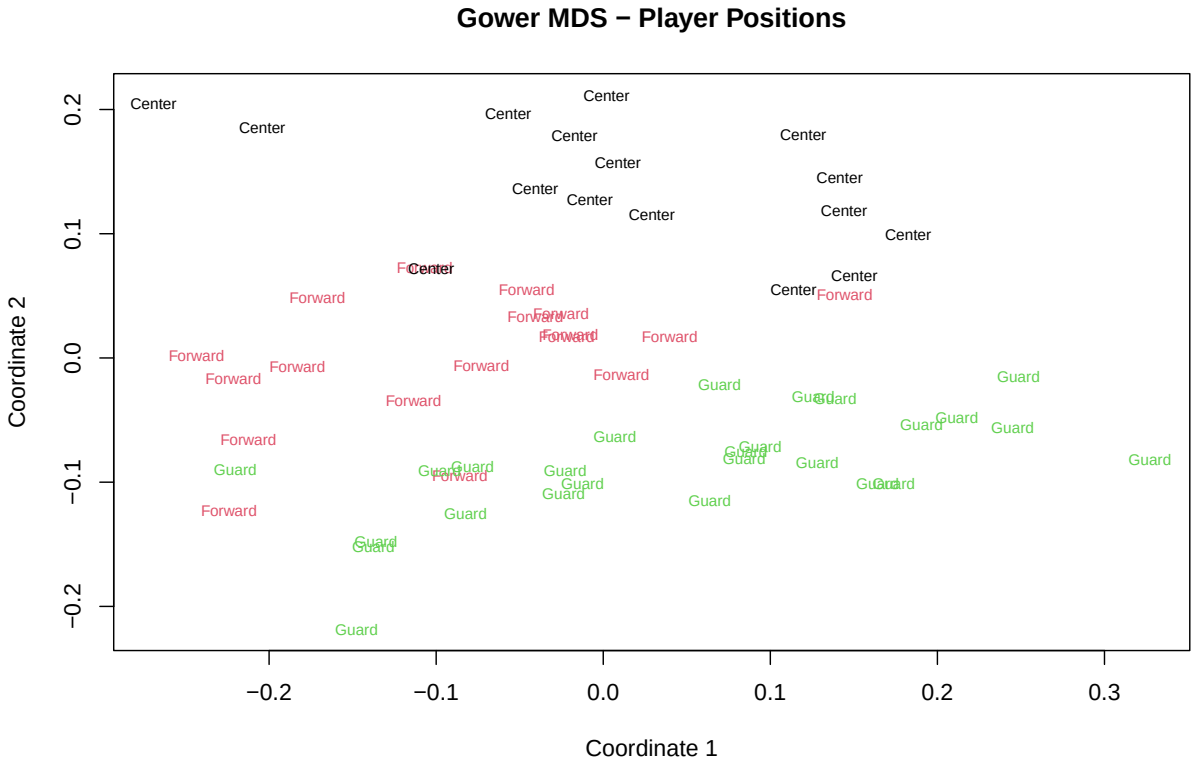
3c. Metric MDS - Gower Distance

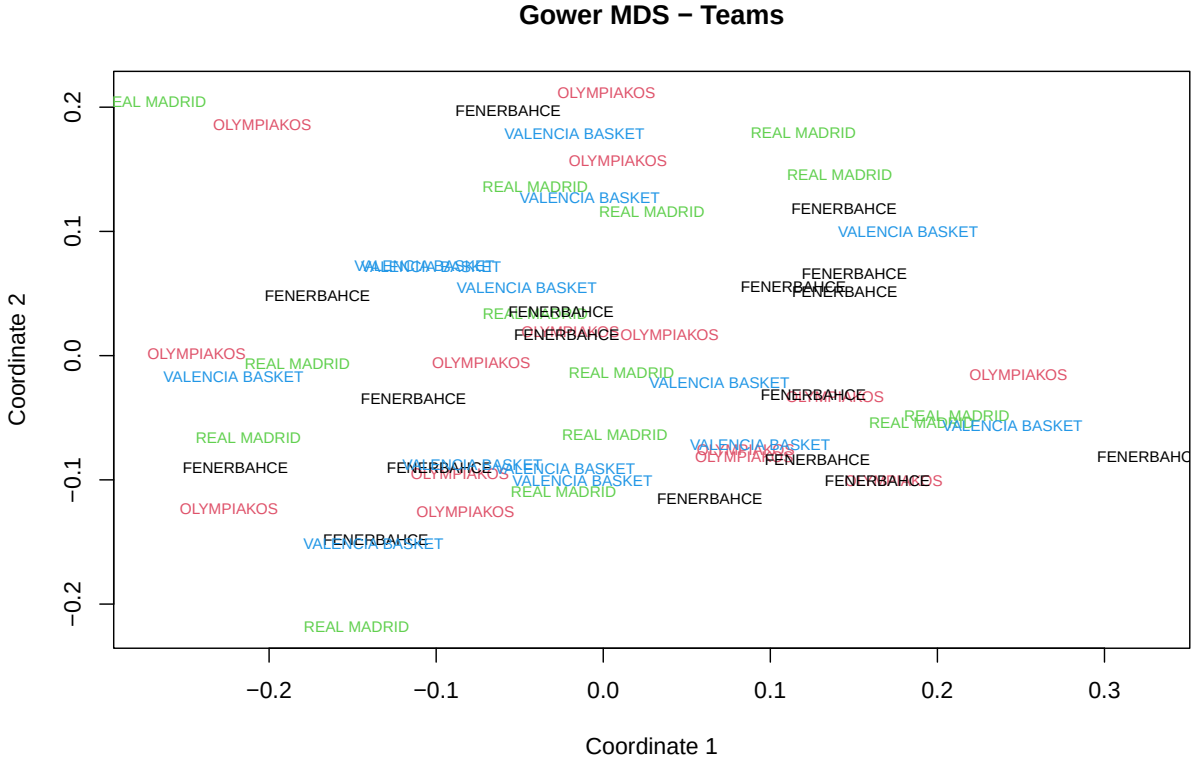


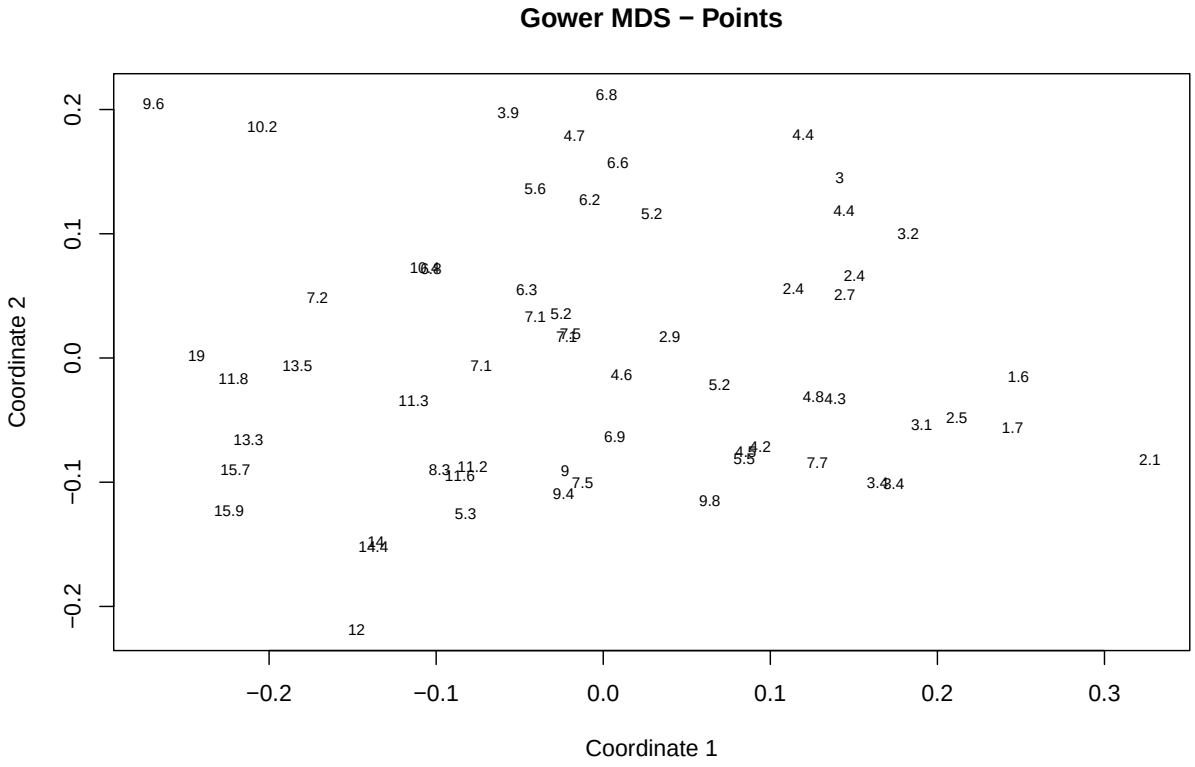

```
##  [1]  1.168329e+00  6.441960e-01  2.886618e-01  1.697698e-01  1.391959e-01
##  [6]  1.135799e-01  1.061880e-01  7.372461e-02  6.500444e-02  5.112237e-02
## [11]  4.733844e-02  4.145983e-02  3.451877e-02  3.114349e-02  2.811408e-02
## [16]  2.458142e-02  2.243850e-02  1.816294e-02  1.734874e-02  1.383134e-02
## [21]  1.329685e-02  8.292776e-03  7.511550e-03  5.477563e-03  4.965205e-03
## [26]  3.008304e-03  2.734736e-03  9.139220e-04 -6.938894e-17 -1.985155e-04
## [31] -7.636607e-04 -1.439072e-03 -2.367760e-03 -4.290621e-03 -4.528827e-03
## [36] -5.504499e-03 -6.442520e-03 -7.685371e-03 -8.305791e-03 -8.546051e-03
## [41] -9.790315e-03 -1.014870e-02 -1.073121e-02 -1.134592e-02 -1.258196e-02
## [46] -1.458524e-02 -1.549999e-02 -1.555963e-02 -1.675239e-02 -1.844129e-02
## [51] -1.952001e-02 -2.235721e-02 -2.620667e-02 -2.666931e-02 -3.098803e-02
## [56] -3.587956e-02 -3.730000e-02 -4.059799e-02 -5.252564e-02 -6.568356e-02
```

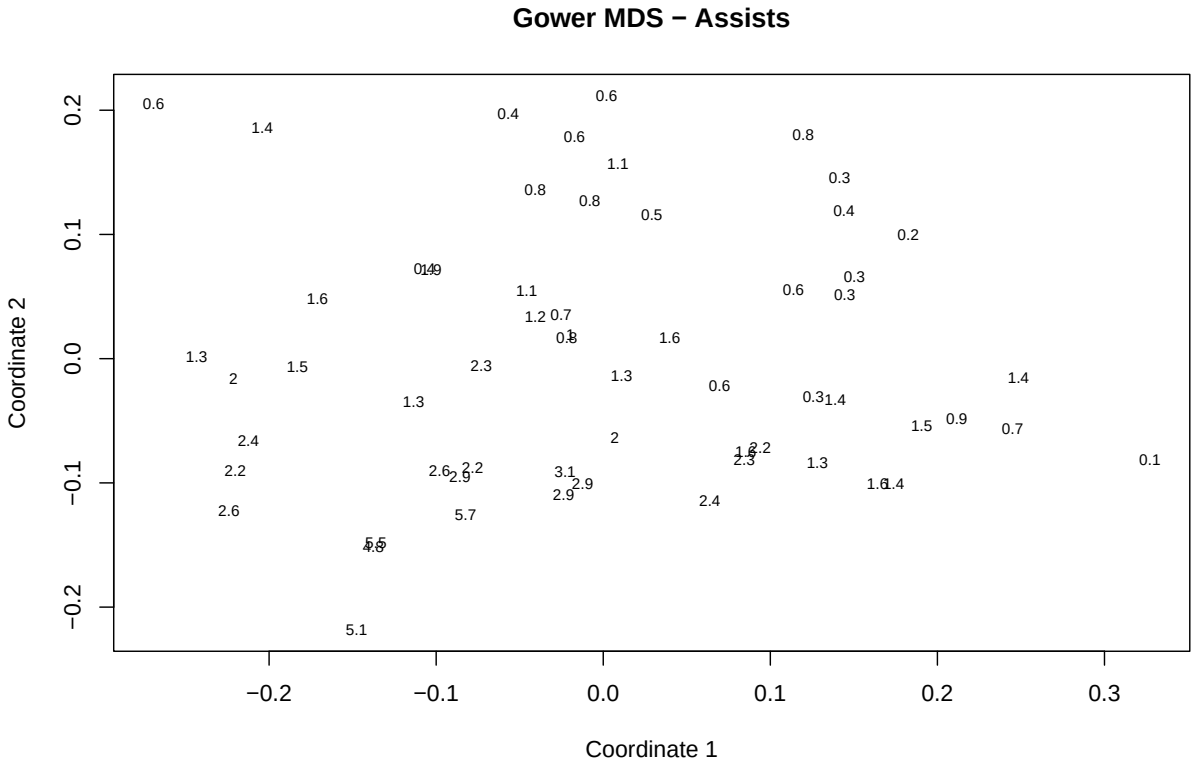
```
## [1] 0.4914459 0.5763360
```

3d. Interpretation of Gower MDS Dimensions









Coordinate 1 – Scoring Activity: Players with higher average points generally appear on the left, while

lower-scoring players tend to appear on the right, suggesting that this dimension partly reflects scoring activity.

Coordinate 2 – Playing Position and Playmaking: Centers are mainly located at positive values, guards at negative values, and forwards in between. Higher assist averages also tend to occur at negative values, indicating that this dimension distinguishes inside-oriented players from playmakers.

Team Membership: Players from all four teams overlap considerably, suggesting that team membership does not strongly explain the MDS dimensions.

3e

Gower MDS provides clearer grouping of players by position, with centers, guards, and forwards showing a more distinct separation. This is because Gower distance incorporates the categorical variable POSITION alongside numerical statistics. Euclidean MDS mainly reflects differences in numerical performance and shows less distinct groups. # 4. Multiple Correspondence Analysis (MCA)

4a. Convert Numerical Variables to Binary Categories

```
# Create MCA dataset
mca_data <- data

# Identify numerical variables
num_vars <- sapply(mca_data, is.numeric)

# Convert numerical variables to binary factors
mca_data[num_vars] <- lapply(mca_data[num_vars], function(x) {
  factor(
    ifelse(x > mean(x, na.rm = TRUE),
           "Over average", "Below average")
  )
})

# Ensure categorical variables are factors
mca_data$TEAM <- as.factor(mca_data$TEAM)
mca_data$POSITION <- as.factor(mca_data$POSITION)

# Inspect results
str(mca_data)
```

```
## 'data.frame':    60 obs. of  20 variables:
##  $ TEAM    : Factor w/ 4 levels "FENERBAHCE","OLYMPIAKOS",..: 2 2 2 2 2 2 2 2 2 2 ...
##  $ POSITION: Factor w/ 3 levels "Center","Forward",..: 3 3 2 3 3 2 2 2 2 1 ...
##  $ GP      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 2 2 2 2 ...
##  $ GS      : Factor w/ 2 levels "Below average",..: 2 1 1 1 1 2 2 2 1 2 ...
##  $ PTS     : Factor w/ 2 levels "Below average",..: 1 1 1 1 1 2 1 2 2 2 ...
##  $ X2P.    : Factor w/ 2 levels "Below average",..: 1 1 2 1 1 2 2 1 2 2 ...
##  $ X3P.    : Factor w/ 2 levels "Below average",..: 2 2 1 2 1 2 2 2 2 1 ...
##  $ FT.     : Factor w/ 2 levels "Below average",..: 2 2 1 2 1 2 1 2 2 2 ...
##  $ OR      : Factor w/ 2 levels "Below average",..: 1 1 1 1 1 2 1 1 1 2 ...
##  $ DR      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 1 2 2 ...
##  $ TR      : Factor w/ 2 levels "Below average",..: 1 1 2 1 1 2 1 1 2 2 ...
##  $ AST     : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 1 1 2 1 1 ...
##  $ STL     : Factor w/ 2 levels "Below average",..: 2 1 1 1 1 2 1 2 1 2 ...
```

```
##  $ TO      : Factor w/ 2 levels "Below average",..: 2 2 2 2 1 2 1 2 1 1 ...
##  $ BLK     : Factor w/ 2 levels "Below average",..: 1 2 1 1 1 1 1 1 1 2 ...
##  $ BLKA    : Factor w/ 2 levels "Below average",..: 1 2 2 2 2 2 1 2 1 2 ...
##  $ FC      : Factor w/ 2 levels "Below average",..: 2 2 2 1 1 1 1 2 2 1 ...
##  $ FD      : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 2 2 ...
##  $ PIR     : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 2 2 ...
##  $ Min 2   : Factor w/ 2 levels "Below average",..: 2 1 2 1 1 2 1 2 1 2 ...
```

```
summary(mca_data)
```

```
##               TEAM       POSITION              GP                  GS
##  FENERBAHCE     :17   Center :16   Below average:21   Below average:37
##  OLYMPIAKOS     :15   Forward:18   Over average :39   Over average :23
##  REAL MADRID    :14   Guard  :26
##  VALENCIA BASKET:14
##             PTS                X2P.               X3P.               FT.
##  Below average:37   Below average:30   Below average:19   Below average:25
##  Over average :23   Over average :30   Over average :41   Over average :35
##
##
##              OR                 DR                 TR                AST
##  Below average:36   Below average:33   Below average:32   Below average:40
##  Over average :24   Over average :27   Over average :28   Over average :20
##
##
##             STL                TO                BLK                BLKA
##  Below average:35   Below average:34   Below average:38   Below average:26
##  Over average :25   Over average :26   Over average :22   Over average :34
##
##
##              FC                 FD                PIR              Min 2
##  Below average:28   Below average:33   Below average:31   Below average:32
##  Over average :32   Over average :27   Over average :29   Over average :28
##
##
```

4b. Multiple Correspondence Analysis

```
##
## Call:
## MCA(X = mca_data, graph = FALSE)
##
##
## Eigenvalues
##                        Dim.1   Dim.2   Dim.3   Dim.4   Dim.5   Dim.6   Dim.7
## Variance               0.293   0.191   0.079   0.074   0.067   0.058   0.051
## % of var.             25.463  16.618   6.883   6.434   5.824   5.028   4.468
## Cumulative % of var.  25.463  42.081  48.964  55.398  61.222  66.250  70.718
##                        Dim.8   Dim.9  Dim.10  Dim.11  Dim.12  Dim.13  Dim.14
## Variance               0.044   0.039   0.035   0.033   0.030   0.028   0.024
## % of var.              3.851   3.401   3.038   2.871   2.608   2.472   2.056
## Cumulative % of var.  74.570  77.970  81.008  83.879  86.487  88.958  91.014
##                       Dim.15  Dim.16  Dim.17  Dim.18  Dim.19  Dim.20  Dim.21
## Variance               0.020   0.018   0.015   0.014   0.012   0.010   0.007
```

```
## % of var.              1.775   1.528   1.307   1.216   1.062   0.838   0.587
## Cumulative % of var.  92.790  94.318  95.625  96.841  97.903  98.741  99.327
##                       Dim.22  Dim.23
## Variance               0.005   0.003
## % of var.              0.405   0.267
## Cumulative % of var.  99.733 100.000
##
## Individuals (the 10 first)
##                         Dim.1    ctr   cos2    Dim.2    ctr   cos2    Dim.3
## THOMAS WALKUP         |  0.423  1.020  0.156 | -0.511  2.276  0.228 | -0.126
## FRANK NTILIKNA        | -0.490  1.365  0.234 | -0.309  0.835  0.094 | -0.100
## TYSON WARD            |  0.433  1.068  0.154 |  0.014  0.002  0.000 | -0.260
## SABEN LEE             | -0.605  2.086  0.374 | -0.460  1.847  0.216 | -0.100
## MONTE MORRIS          | -0.763  3.314  0.544 | -0.138  0.167  0.018 |  0.010
## SASHA VEZENKOV        |  0.777  3.434  0.497 |  0.013  0.002  0.000 | -0.275
## KOSTAS PAPANIKOLAU    | -0.420  1.004  0.168 |  0.063  0.034  0.004 | -0.052
## TYLER DORSEY          |  0.572  1.865  0.273 | -0.694  4.203  0.402 | -0.118
## ALEC PETERS           |  0.207  0.243  0.040 |  0.097  0.082  0.009 | -0.640
## NIKOLA MILUTINOV      |  0.658  2.464  0.321 |  0.486  2.058  0.175 | -0.167
##                         ctr   cos2
## THOMAS WALKUP         0.334  0.014 |
## FRANK NTILIKNA        0.210  0.010 |
## TYSON WARD            1.418  0.055 |
## SABEN LEE             0.209  0.010 |
## MONTE MORRIS          0.002  0.000 |
## SASHA VEZENKOV        1.595  0.062 |
## KOSTAS PAPANIKOLAU    0.058  0.003 |
## TYLER DORSEY          0.294  0.012 |
## ALEC PETERS           8.611  0.386 |
## NIKOLA MILUTINOV      0.584  0.021 |
##
## Categories (the 10 first)
##                          Dim.1    ctr   cos2 v.test    Dim.2    ctr   cos2
## FENERBAHCE             | -0.202  0.197  0.016 -0.974 | -0.081  0.049  0.003
## OLYMPIAKOS             |  0.020  0.002  0.000  0.088 | -0.159  0.164  0.008
## REAL MADRID            |  0.028  0.003  0.000  0.118 |  0.172  0.181  0.009
## VALENCIA BASKET        |  0.196  0.153  0.012  0.829 |  0.097  0.057  0.003
## Center                 | -0.181  0.148  0.012 -0.836 |  1.127  8.856  0.462
## Forward                |  0.548  1.540  0.129  2.757 |  0.073  0.042  0.002
## Guard                  | -0.268  0.533  0.055 -1.803 | -0.744  6.275  0.423
## GP_Below average       | -1.004  6.019  0.542 -5.656 | -0.175  0.280  0.016
## GP_Over average        |  0.540  3.241  0.542  5.656 |  0.094  0.151  0.016
## GS_Below average       | -0.527  2.924  0.447 -5.134 |  0.024  0.009  0.001
##                        v.test    Dim.3    ctr   cos2 v.test
## FENERBAHCE             -0.393 | -0.834 12.463  0.275 -4.030 |
## OLYMPIAKOS             -0.703 | -0.688  7.466  0.158 -3.049 |
## REAL MADRID             0.729 |  0.561  4.639  0.096  2.377 |
## VALENCIA BASKET         0.409 |  1.189 20.837  0.430  5.039 |
## Center                  5.219 |  0.254  1.084  0.023  1.175 |
## Forward                 0.368 | -0.545  5.625  0.127 -2.740 |
## Guard                  -4.997 |  0.221  1.337  0.037  1.485 |
## GP_Below average       -0.986 | -0.230  1.172  0.029 -1.298 |
## GP_Over average         0.986 |  0.124  0.631  0.029  1.298 |
## GS_Below average        0.232 | -0.185  1.332  0.055 -1.802 |
```

```
##
## Categorical variables (eta2)
##                         Dim.1 Dim.2 Dim.3
## TEAM                  | 0.021 0.017 0.719 |
## POSITION              | 0.130 0.580 0.127 |
## GP                    | 0.542 0.016 0.029 |
## GS                    | 0.447 0.001 0.055 |
## PTS                   | 0.564 0.134 0.004 |
## X2P.                  | 0.047 0.287 0.005 |
## X3P.                  | 0.002 0.161 0.017 |
## FT.                   | 0.042 0.215 0.224 |
## OR                    | 0.128 0.555 0.026 |
## DR                    | 0.423 0.323 0.035 |
```

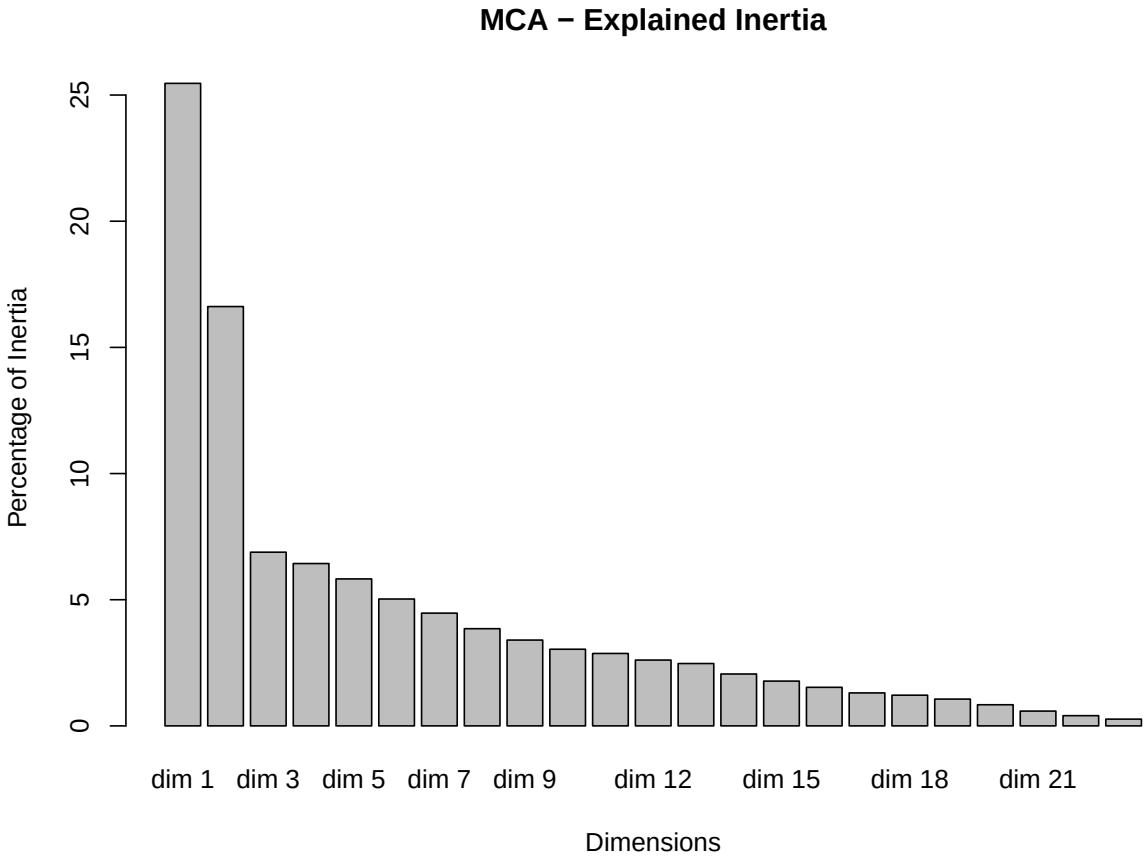
```
##         eigenvalue percentage of variance cumulative percentage of variance
## dim 1  0.292824299             25.4629826                             25.46298
## dim 2  0.191107493             16.6180429                             42.08103
## dim 3  0.079159636              6.8834466                             48.96447
## dim 4  0.073986556              6.4336136                             55.39809
## dim 5  0.066973649              5.8237955                             61.22188
## dim 6  0.057825182              5.0282767                             66.25016
## dim 7  0.051381465              4.4679535                             70.71811
## dim 8  0.044291771              3.8514584                             74.56957
## dim 9  0.039110117              3.4008797                             77.97045
## dim 10 0.034932279              3.0375895                             81.00804
## dim 11 0.033011571              2.8705714                             83.87861
## dim 12 0.029994779              2.6082417                             86.48685
## dim 13 0.028422678              2.4715372                             88.95839
## dim 14 0.023644865              2.0560752                             91.01446
## dim 15 0.020413387              1.7750771                             92.78954
## dim 16 0.017576632              1.5284028                             94.31794
## dim 17 0.015029621              1.3069236                             95.62487
## dim 18 0.013987156              1.2162744                             96.84114
## dim 19 0.012213189              1.0620164                             97.90316
## dim 20 0.009634213              0.8377577                             98.74092
## dim 21 0.006745394              0.5865560                             99.32747
## dim 22 0.004658046              0.4050475                             99.73252
## dim 23 0.003076020              0.2674800                            100.00000
```

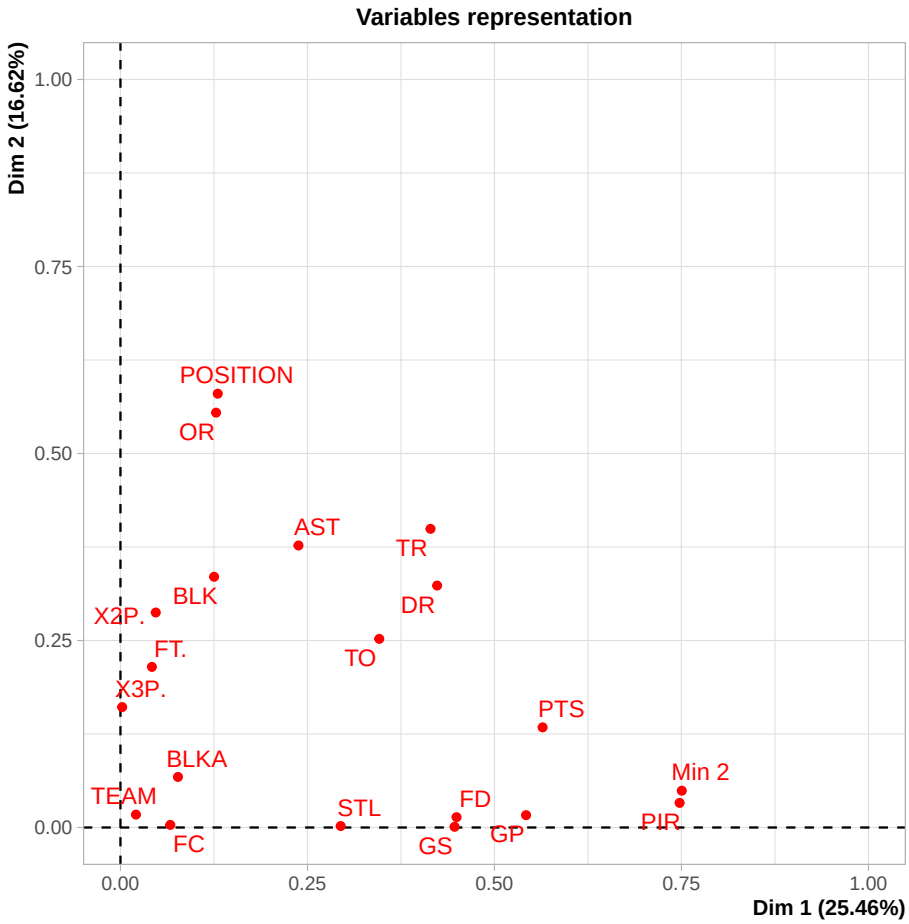
```
## $`Dim 1`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                  R2      p.value
## Min 2    0.75049087 3.924760e-19
## PIR      0.74759117 5.497210e-19
## PTS      0.56439705 4.685675e-12
## GP       0.54230206 2.004974e-11
## FD       0.44935765 4.664512e-09
## GS       0.44674418 5.365644e-09
## DR       0.42338548 1.825151e-08
## TR       0.41456907 2.861885e-08
## TO       0.34606063 7.701888e-07
## STL      0.29445518 7.501258e-06
## AST      0.23804330 7.678147e-05
```

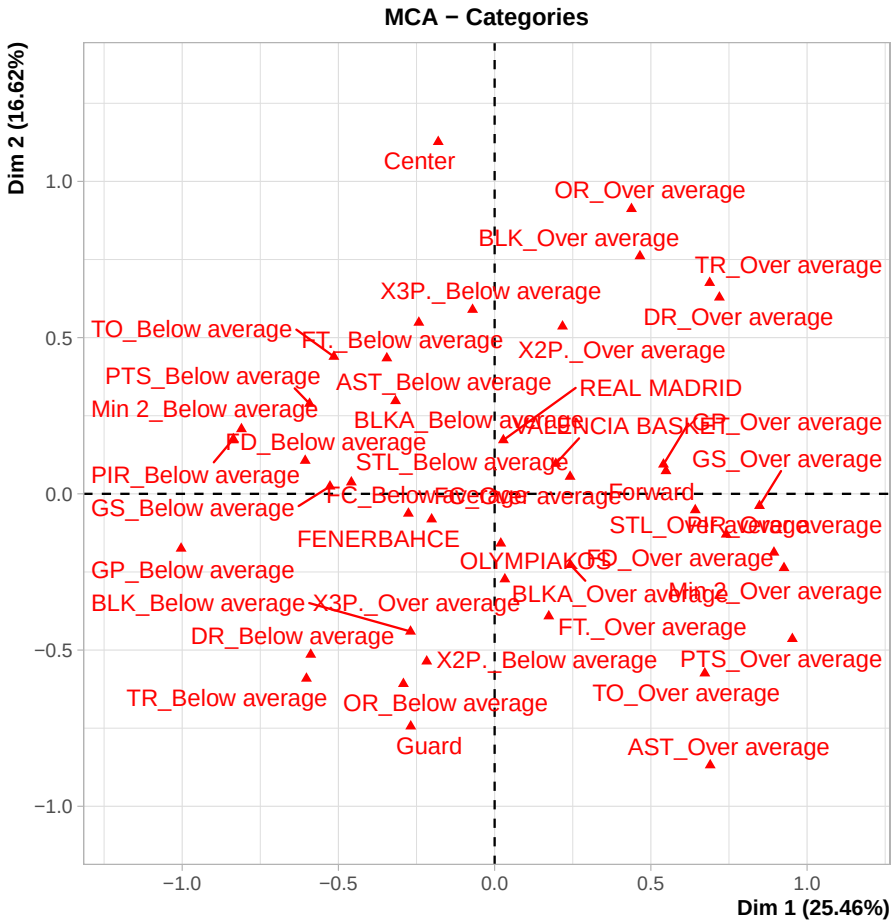
```
## OR       0.12787760 5.030215e-03
## BLK      0.12521652 5.542554e-03
## POSITION 0.13009925 1.883157e-02
## BLKA     0.07690254 3.194181e-02
## FC       0.06666730 4.638722e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                              Estimate      p.value
## Min 2=Min 2_Over average   0.4698330 3.924760e-19
## PIR=PIR_Over average       0.4681414 5.497210e-19
## PTS=PTS_Over average       0.4180733 4.685675e-12
## GP=GP_Over average         0.4177375 2.004974e-11
## FD=FD_Over average         0.3645709 4.664512e-09
## GS=GS_Over average         0.3719542 5.365644e-09
## DR=DR_Over average         0.3538783 1.825151e-08
## TR=TR_Over average         0.3491960 2.861885e-08
## TO=TO_Over average         0.3211994 7.701888e-07
## STL=STL_Over average       0.2978039 7.501258e-06
## AST=AST_Over average       0.2800321 7.678147e-05
## POSITION=Forward           0.2787756 4.859760e-03
## OR=OR_Over average         0.1974991 5.030215e-03
## BLK=BLK_Over average       0.1986792 5.542554e-03
## BLKA=BLKA_Over average     0.1514150 3.194181e-02
## FC=FC_Over average         0.1400320 4.638722e-02
## FC=FC_Below average       -0.1400320 4.638722e-02
## BLKA=BLKA_Below average   -0.1514150 3.194181e-02
## BLK=BLK_Below average     -0.1986792 5.542554e-03
## OR=OR_Below average       -0.1974991 5.030215e-03
## AST=AST_Below average     -0.2800321 7.678147e-05
## STL=STL_Below average     -0.2978039 7.501258e-06
## TO=TO_Below average       -0.3211994 7.701888e-07
## TR=TR_Below average       -0.3491960 2.861885e-08
## DR=DR_Below average       -0.3538783 1.825151e-08
## GS=GS_Below average       -0.3719542 5.365644e-09
## FD=FD_Below average       -0.3645709 4.664512e-09
## GP=GP_Below average       -0.4177375 2.004974e-11
## PTS=PTS_Below average     -0.4180733 4.685675e-12
## PIR=PIR_Below average     -0.4681414 5.497210e-19
## Min 2=Min 2_Below average -0.4698330 3.924760e-19
##
## $`Dim 2`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                   R2      p.value
## OR       0.55460719 9.000731e-12
## POSITION 0.57992609 1.839852e-11
## TR       0.39928150 6.149994e-08
## AST      0.37703318 1.812981e-07
## BLK      0.33520399 1.259809e-06
## DR       0.32344402 2.129237e-06
## X2P.     0.28739466 1.012286e-05
## TO       0.25219391 4.345197e-05
```

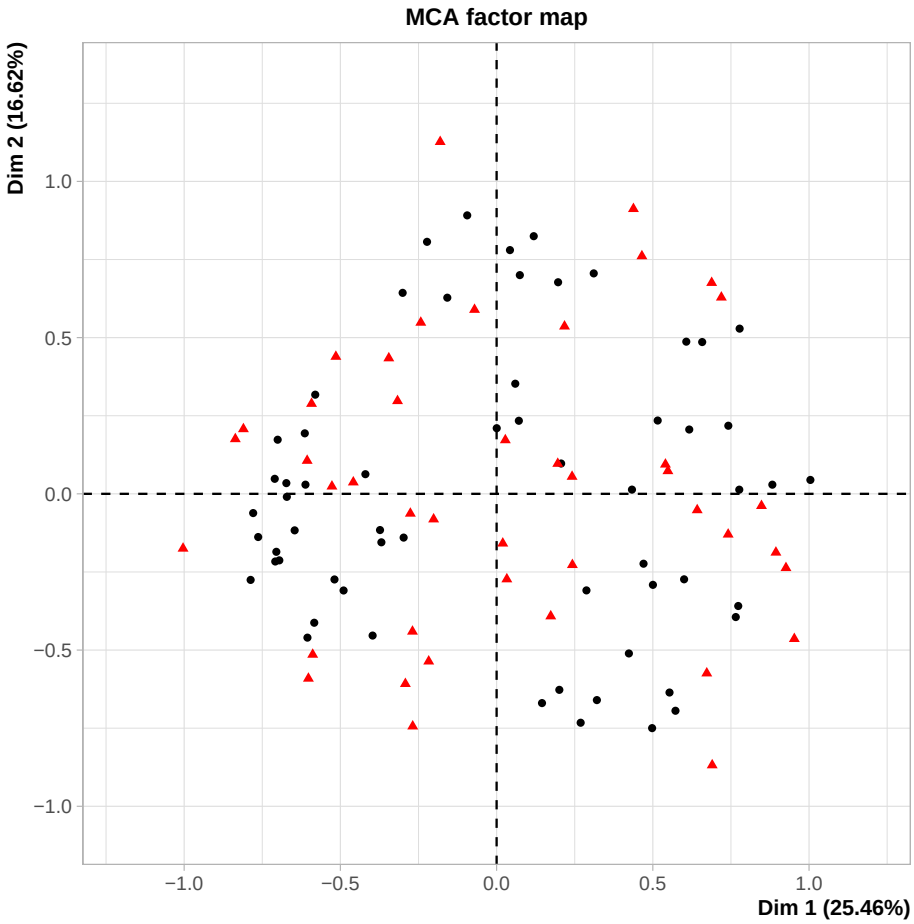
```
## FT.      0.21469340 1.928376e-04
## X3P.     0.16097706 1.487605e-03
## PTS      0.13389107 4.038357e-03
## BLKA     0.06759126 4.484562e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                            Estimate      p.value
## OR=OR_Over average        0.3322737 9.000731e-12
## POSITION=Center           0.4260985 2.403476e-09
## TR=TR_Over average        0.2768507 6.149994e-08
## AST=AST_Below average     0.2847114 1.812981e-07
## BLK=BLK_Over average      0.2626102 1.259809e-06
## DR=DR_Over average        0.2498739 2.129237e-06
## X2P.=X2P._Over average    0.2343571 1.012286e-05
## TO=TO_Below average       0.2215140 4.345197e-05
## FT.=FT._Below average     0.2054307 1.928376e-04
## X3P.=X3P._Below average   0.1885269 1.487605e-03
## PTS=PTS_Below average     0.1645020 4.038357e-03
## BLKA=BLKA_Below average   0.1146778 4.484562e-02
## BLKA=BLKA_Over average   -0.1146778 4.484562e-02
## PTS=PTS_Over average     -0.1645020 4.038357e-03
## X3P.=X3P._Over average   -0.1885269 1.487605e-03
## FT.=FT._Over average     -0.2054307 1.928376e-04
## TO=TO_Over average       -0.2215140 4.345197e-05
## X2P.=X2P._Below average -0.2343571 1.012286e-05
## DR=DR_Below average      -0.2498739 2.129237e-06
## BLK=BLK_Below average    -0.2626102 1.259809e-06
## AST=AST_Over average     -0.2847114 1.812981e-07
## TR=TR_Below average      -0.2768507 6.149994e-08
## POSITION=Guard           -0.3916480 1.839051e-08
## OR=OR_Below average      -0.3322737 9.000731e-12
##
## $`Dim 3`
##
## Link between the variable and the categorical variable (1-way anova)
## ============================================
##                   R2      p.value
## TEAM     0.71884776 1.920278e-15
## FT.      0.22362755 1.359259e-04
## FD       0.12369051 5.859276e-03
## POSITION 0.12738633 2.057915e-02
## BLKA     0.07778667 3.093214e-02
## AST      0.06905669 4.250729e-02
##
## Link between variable and the categories of the categorical variables
## ============================================================
##                              Estimate      p.value
## TEAM=VALENCIA BASKET     0.31850600 1.277497e-08
## FT.=FT._Below average    0.13493722 1.359259e-04
## FD=FD_Below average      0.09944947 5.859276e-03
## TEAM=REAL MADRID         0.14181455 1.611014e-02
## BLKA=BLKA_Over average   0.07917710 3.093214e-02
## AST=AST_Over average     0.07842076 4.250729e-02
```

```
## AST=AST_Below average   -0.07842076 4.250729e-02
## BLKA=BLKA_Below average -0.07917710 3.093214e-02
## FD=FD_Over average      -0.09944947 5.859276e-03
## POSITION=Forward        -0.14671615 5.154202e-03
## TEAM=OLYMPIAKOS         -0.20949884 1.685833e-03
## FT.=FT._Over average    -0.13493722 1.359259e-04
## TEAM=FENERBAHCE         -0.25082170 1.680275e-05
```




Variables representation
Dim 2 (16.62%)
1.00
0.75
0.50
0.25
0.00
POSITION
OR
AST
TR
BLK
DR
X2P.
FT.
TO
X3P.
PTS
BLKA
Min 2
TEAM
STL
FD
PIR
FC
GS
GP
0.00
0.25
0.50
0.75
1.00
Dim 1 (25.46%)


MCA − Categories
Dim 2 (16.62%)
Dim 1 (25.46%)
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Center
OR_Over average
BLK_Over average
TR_Over average
X3P._Below average
DR_Over average
TO_Below average
FT._Below average
X2P._Over average
PTS_Below average
AST_Below average
REAL MADRID
Min 2_Below average
BLKA_Below average
VALENCIA BASKET
GP_Over average
FD_Below average
STL_Below average
GS_Over average
PIR_Below average
FC_Below average
Forward
GS_Below average
STL_Over average
FENERBAHCE
OLYMPIAKOS
FD_Over average
GP_Below average
BLK_Below average
X3P._Over average
BLKA_Over average
Min 2_Over average
FT._Over average
DR_Below average
X2P._Below average
PTS_Over average
TR_Below average
OR_Below average
TO_Over average
Guard
AST_Over average



The first two MCA dimensions explain **42.08% of the total inertia** (25.46% and 16.62%).

Dimension 1 – Overall Statistical Involvement: This dimension is mainly associated with minutes played ($R^2 = 0.750$), PIR ($R^2 = 0.748$), and points scored ($R^2 = 0.564$). Above-average performance categories are located on the positive side, while below-average categories appear on the negative side.

Dimension 2 – Inside Play vs. Playmaking: Playing position ($R^2 = 0.580$), offensive rebounds ($R^2 = 0.555$), and assists ($R^2 = 0.377$) strongly characterize this dimension. Centers and above-average rebounding categories are located at positive values, while guards and above-average assists appear at negative values.

The individual factor map shows differences in player profiles without clearly separated clusters. Team categories are located near the origin, suggesting little influence on the first two dimensions. However, Dimension 3 (6.88%) is strongly associated with team membership ($R^2 = 0.719$).

4c

All three methods reveal similar patterns. PCA and MCA identify overall statistical involvement as the first dimension and differences between playmaking and inside-oriented players as the second. Euclidean MDS highlights differences in numerical player statistics, while Gower MDS provides clearer separation by playing position. Unlike PCA, MCA uses categorized performance variables, resulting in some information loss but making above- and below-average performance profiles easier to interpret.